Function Exercises and Inequality Estimates

N-20221105

§1.1 A short note about elementary methods

A person who understands the structure of a problem sees more than someone who only imitates formulas. The following exercise sheet develops that principle through periodicity, quadratic minima, logarithmic comparison, parameter ranges, and an inequality with $abc = 1$, with the reasoning made explicit.

§1.2 Periodicity from a transformed argument

The first problem has a function f satisfying a relation of the form

$$f(x+4) = f(x), \qquad g(x+4) = g(x) + 2,$$

and asks about the periodicity of a composite or transformed function. The increment in x produces the same value of f and a controlled increment of g. The general principle is:

$$f(x+T) = f(x) \quad \Longrightarrow \quad f \text{ has period } T.$$

If a formula contains $x+8$, use the period twice. For example,

$$f(x+8) = f((x+4)+4) = f(x+4) = f(x).$$

The important point is that a period is not necessarily the least positive period. A function with period 4 automatically has period $8, 12, \ldots$. Many errors come from confusing "a period" with "the fundamental period."

§1.3 A quadratic minimum

The second problem studies

$$y = x^2 + x + \frac{5}{3}.$$

Complete the square:

$$y = \left(x + \frac{1}{2}\right)^2 + \frac{17}{12}.$$

Hence

$$y \geq \frac{17}{12},$$

with equality at

$$x = -\frac{1}{2}.$$

This is the cleanest form of a quadratic problem. Differentiation gives the same result, but completing the square better reveals the geometry: a parabola shifted horizontally and vertically.

§1.4 A logarithmic comparison

Another problem compares functions involving logarithms, for example

$$f(x) = \log_a(x+1)$$

and a related expression $g(x) - f(x)$. Logarithm rules rewrite the difference as:

$$\log_a u - \log_a v = \log_a \frac{u}{v}.$$

Then the sign of the difference depends on whether the base a is greater than 1 or between 0 and 1. This is the crucial trap:

$$a > 1 \quad \Rightarrow \quad \log_a x \text{ is increasing,}$$

whereas

$$0 < a < 1 \quad \Rightarrow \quad \log_a x \text{ is decreasing.}$$

So a logarithmic inequality must always track the base.

§1.5 A parameterized polynomial

For a parameter p and a function with a specified zero or extremum, substitute the relevant value and reduce the condition to an equation in p. The broad template is:

(i) if x_0 is a zero, impose $f(x_0) = 0$;

(ii) if x_0 is an extremum of a differentiable function, impose $f'(x_0) = 0$;

(iii) if a graph is tangent to a line, impose both value equality and slope equality.

This page is mainly training the habit of translating a verbal condition into equations.

§1.6 An inequality under $abc = 1$

The last problem states: for $a, b, c > 0$, $abc = 1$, prove an inequality of the form

$$\frac{1}{a^3(cb+c)} + \frac{1}{b^3(ca+c)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

Use the condition

$$abc = 1$$

to replace reciprocal expressions by symmetric sums such as

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = ab + bc + ca.$$

Then Cauchy–Schwarz or AM–GM gives a lower bound.

The useful lesson is that the constraint $abc = 1$ should not sit unused. It usually allows conversion between variables and reciprocals. A standard move is also

$$a = \frac{x}{y}, \qquad b = \frac{y}{z}, \qquad c = \frac{z}{x},$$

which automatically enforces $abc = 1$. Whether one uses this substitution or direct reciprocal rewriting depends on the shape of the expression.

§1.7 The conceptual turn

The exercises look miscellaneous, but the same strategy recurs: normalize the expression before attacking it. Periodicity reduces arguments modulo a period; completing the square normalizes a quadratic; logarithm laws turn products into sums; parameter conditions become equations; $abc = 1$ turns reciprocal expressions into symmetric ones. Elementary problem solving is often the art of finding the right normalization.

§1.8 Periodicity from transformed arguments

A relation such as $f(x+a) = f(x)$ is invariance under translation by a. More complicated relations, for example involving $f(c-x)$, combine translation and reflection. The group generated by these transformations controls the function's repeated values.

Thus many functional-equation exercises are really symmetry problems on the domain.

§1.9 Quadratic and logarithmic estimates

A quadratic minimum is controlled by completing the square:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

A logarithmic comparison often uses monotonicity or concavity of log. In both cases, the point is to expose the structural feature: convexity for quadratics, monotone concavity for logarithms.

§1.10 Functional equations and orbit reduction

If a functional equation lets one transform x into simpler inputs, then the domain splits into orbits under the generated transformations. A fundamental domain is a set of representatives for these orbits.

Solving the equation means assigning values on the fundamental domain and transporting them consistently along the orbit.

§1.11 Functions through symmetry and normal form

Function exercises are unified by structure: periodicity is group invariance, quadratics are convex normal forms, logarithmic inequalities come from monotonicity and concavity, and functional equations reduce values along orbits.